

CLOUD COMPUTING

UE23CS351B

LAB 4- Docker Advanced: Networks, Volumes, and Microservices

GITHUB:

**[https://github.com/akshayag2005/CLOUD-
COMPUTING](https://github.com/akshayag2005/CLOUD-COMPUTING)**

DATE: 02/02/2026

NAME: AKSHAY

SRN: PES2UG23CS044

SECTION: A

SS1:

```
[akshay@AKSHAY]~$ export SRN=PES2UG23CS044

[akshay@AKSHAY]~$ echo $SRN
PES2UG23CS044

[akshay@AKSHAY]~$ export PS1="\u@$SRN \W]\$ "

[akshay@PES2UG23CS044 ~]$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

[akshay@PES2UG23CS044 ~]$
```

SS2:

```
[akshay@PES2UG23CS044 ~]$ docker-compose --version
Docker Compose version v5.0.1

[akshay@PES2UG23CS044 ~]$ |
```

SS3:

```
[akshay@PES2UG23CS044 docker-advanced-lab]$ docker run -d --name rabbitmq rabbitmq:3-management
89ac47c86593720c1b381dd8c3ba0a2f41fce913dd782a237727ad22716ac7f2

[akshay@PES2UG23CS044 docker-advanced-lab]$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
89ac47c86593   rabbitmq:3-management              "docker-entrypoint.s..." 27 seconds ago Up 26 seconds 4369/tcp, 5671-5672/tcp, 15671-15672/tcp, 15691-15692/tcp, 25672/tcp
               rabbitmq

[akshay@PES2UG23CS044 docker-advanced-lab]$ |
```

SS4:

```
[akshay@PES2UG23CS044 docker-advanced-lab]$ docker run -it --name user-a chat-app

--- No Previous Chat History ---

--- Chat Started (unknown-user connecting to localhost) ---
RabbitMQ not ready, retrying... (1/5)
RabbitMQ not ready, retrying... (1/5)
RabbitMQ not ready, retrying... (2/5)
RabbitMQ not ready, retrying... (2/5)
RabbitMQ not ready, retrying... (3/5)
RabbitMQ not ready, retrying... (3/5)
RabbitMQ not ready, retrying... (4/5)
RabbitMQ not ready, retrying... (4/5)
RabbitMQ not ready, retrying... (5/5)
RabbitMQ not ready, retrying... (5/5)

ERROR: Could not connect to RabbitMQ at localhost

ERROR: Could not connect to RabbitMQ at localhost

Falling back to standalone mode...
=====
STANDALONE MODE (No RabbitMQ)
=====

--- No Previous Chat History ---

Messages will not be sent to any other container.

Type message (standalone): |
```

SS5:

```
[akshay@PES2UG23CS044 docker-advanced-lab]$ docker run -it --name user-a \
--network chat-net \
-e RABBIT_HOST=rabbitmq \
-e QUEUE_NAME=queue_a \
-e TARGET_QUEUE=queue_b \
chat-app

--- No Previous Chat History ---

--- Chat Started (unknown-user connecting to rabbitmq) ---
Type message: hello
Sent by unknown-user: hello
Type message:
Received from unknown-user: hello friend
Type message: hope you're enjoying the lab :)
Sent by unknown-user: hope you're enjoying the lab :)
Type message:
Received from unknown-user: no It's pretty boring!
Type message: |
```

SS6:

```
(akshay@AKSHAY) - [/mnt/c/Users/AKSHAY_AG]
$ docker run -it --name user-b \
  --network chat-net \
  -e RABBIT_HOST=rabbitmq \
  -e QUEUE_NAME=queue_b \
  -e TARGET_QUEUE=queue_a \
  chat-app

--- No Previous Chat History ---

--- Chat Started (unknown-user connecting to rabbitmq) ---
Type message:
Received from unknown-user: hello
Type message: hello friend
Sent by unknown-user: hello friend
Type message:
Received from unknown-user: hope you're enjoying the lab :)
Type message: no It's pretty boring!
Sent by unknown-user: no It's pretty boring!
Type message: |
```

SS7:

```
[akshay@PES2UG23CS044 docker-advanced-lab]$ docker stop user-a user-b rabbitmq
user-a
user-b
rabbitmq

[akshay@PES2UG23CS044 docker-advanced-lab]$ docker rm user-a user-b rabbitmq
user-a
user-b
rabbitmq

[akshay@PES2UG23CS044 docker-advanced-lab]$ docker network rm chat-net
chat-net
```

SS8:

```
rabbitmq-1 | 2026-02-05 16:52:35.586285+00:00 [info] <0.254.0> Running boot step rabbit_quorum_queue_periodic_membership_reconciliation defined by app rabbit
rabbitmq-1 | 2026-02-05 16:52:35.586755+00:00 [info] <0.254.0> Running boot step definition_import_worker_pool defined by app rabbit
rabbitmq-1 | 2026-02-05 16:52:35.586955+00:00 [info] <0.315.0> Starting worker pool 'definition_import_pool' with 16 processes in it
rabbitmq-1 | 2026-02-05 16:52:35.589038+00:00 [info] <0.254.0> Running boot step cluster_name defined by app rabbit
rabbitmq-1 | 2026-02-05 16:52:35.589727+00:00 [info] <0.254.0> Initialising internal cluster ID to 'rabbitmq-cluster-id-szT7S3vUEUvtN5s4eCX7MQ'
rabbitmq-1 | 2026-02-05 16:52:35.596450+00:00 [info] <0.254.0> Running boot step virtual_host_reconciliation defined by app rabbit
rabbitmq-1 | 2026-02-05 16:52:35.597124+00:00 [info] <0.254.0> Running boot step direct_client defined by app rabbit
rabbitmq-1 | 2026-02-05 16:52:35.597805+00:00 [info] <0.254.0> Running boot step rabbit_federation_exchange defined by app rabbitmq_federation
rabbitmq-1 | 2026-02-05 16:52:35.598251+00:00 [info] <0.254.0> Running boot step rabbit_management_load_definitions defined by app rabbitmq_management
rabbitmq-1 | 2026-02-05 16:52:35.598859+00:00 [info] <0.704.0> Resetting node maintenance status
rabbitmq-1 | 2026-02-05 16:52:35.983890+00:00 [warning] <0.733.0> Deprecated features: 'management_metrics_collection': Feature 'management_metrics_collection' is deprecated.
rabbitmq-1 | 2026-02-05 16:52:35.983890+00:00 [warning] <0.733.0> By default, this feature can still be used for now.
rabbitmq-1 | 2026-02-05 16:52:35.983890+00:00 [warning] <0.733.0> Its use will not be permitted by default in a future minor RabbitMQ version and the feature will be removed from a future major RabbitMQ version; actual versions to be determined.
rabbitmq-1 | 2026-02-05 16:52:35.983890+00:00 [warning] <0.733.0> To continue using this feature when it is not permitted by default, set the following parameter in your configuration:
rabbitmq-1 | 2026-02-05 16:52:35.983890+00:00 [warning] <0.733.0> "deprecated_features.permit.management_metrics_collection = true"
rabbitmq-1 | 2026-02-05 16:52:35.983890+00:00 [warning] <0.733.0> To test RabbitMQ as if the feature was removed, set this in your configuration:
rabbitmq-1 | 2026-02-05 16:52:35.983890+00:00 [warning] <0.733.0> "deprecated_features.permit.management_metrics_collection = false"
rabbitmq-1 | 2026-02-05 16:52:45.564865+00:00 [info] <0.770.0> Management plugin: HTTP (non-TLS) listener started on port 15672
rabbitmq-1 | 2026-02-05 16:52:45.565670+00:00 [info] <0.800.0> Statistics database started.
rabbitmq-1 | 2026-02-05 16:52:45.565957+00:00 [info] <0.799.0> Starting worker pool 'management_worker_pool' with 3 processes in it
rabbitmq-1 | 2026-02-05 16:52:45.589973+00:00 [info] <0.818.0> Prometheus metrics: HTTP (non-TLS) listener started on port 15692
rabbitmq-1 | 2026-02-05 16:52:45.590459+00:00 [info] <0.704.0> Ready to start client connection listeners
rabbitmq-1 | 2026-02-05 16:52:45.596859+00:00 [info] <0.862.0> started TCP listener on [::]:5672
rabbitmq-1 | 2026-02-05 16:52:45.814591+00:00 [info] <0.704.0> completed with 5 plugins.
rabbitmq-1 | 2026-02-05 16:52:45.814591+00:00 [info] <0.704.0> Server startup complete; 5 plugins started.
rabbitmq-1 | 2026-02-05 16:52:45.814591+00:00 [info] <0.704.0> * rabbitmq_prometheus
rabbitmq-1 | 2026-02-05 16:52:45.814591+00:00 [info] <0.704.0> * rabbitmq_federation
rabbitmq-1 | 2026-02-05 16:52:45.814591+00:00 [info] <0.704.0> * rabbitmq_management
rabbitmq-1 | 2026-02-05 16:52:45.814591+00:00 [info] <0.704.0> * rabbitmq_management_agent
rabbitmq-1 | 2026-02-05 16:52:45.814591+00:00 [info] <0.704.0> * rabbitmq_web_dispatch
rabbitmq-1 | 2026-02-05 16:52:45.863623+00:00 [info] <0.9.0> Time to start RabbitMQ: 21252 ms

View in Docker Desktop View Config Enable Watch Detach
```

SS9:

```
=> => transferring context: 2B 0.0s
=> [1/5] FROM docker.io/library/python:3.9--slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf1731b1b 0.1s
=> => resolve docker.io/library/python:3.9--slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf1731b1b 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 4.17kB 0.0s
=> CACHED [2/5] RUN pip install pika 0.0s
=> CACHED [3/5] WORKDIR /app 0.0s
=> CACHED [4/5] COPY chat.py . 0.0s
=> CACHED [5/5] RUN mkdir -p /app/data 0.0s
=> exporting to image 0.3s
=> => exporting layers 0.0s
=> => exporting manifest sha256:3f139ce2af04708739d4cead234aa95f50c5b9a35c34c2afc50d352c2f06a391 0.0s
=> => exporting config sha256:6acd8e79a11e331801f48f4c1a1b2d302e43fb99c093d8874428fc2834fdbc78 0.0s
=> => exporting attestation manifest sha256:f3d082c9f5e383b8bc6a18ec35ef29ac08ab296c04fdef0fbaffe6a39feddb53 0.1s
=> => exporting manifest list sha256:623ab708450331ba249d7ceebc19ecb8869663776fb6597f583a670ce78bd 0.0s
=> => naming to docker.io/library/docker-advanced-lab-user-a:latest 0.0s
=> => unpacking to docker.io/library/docker-advanced-lab-user-a:latest 0.1s
=> resolving provenance for metadata file 0.0s
Image docker-advanced-lab-user-a Built
Container docker-advanced-lab-user-a-run-9821a2c4688e Creating
Container docker-advanced-lab-user-a-run-9821a2c4688e Created

--- No Previous Chat History ---

--- Chat Started (user-a connecting to rabbitmq) ---
Type message: hello
Sent by user-a: hello
Type message:
Received from user-b: wazzzzzz
Type message: wazzzzaaaa
Sent by user-a: wazzzzaaaa
Type message:
```

SS10:

```
=> => transferring context: 28 0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf1731b1b 0.1s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf1731b1b 0.1s
=> [internal] load build context 0.0s
=> => transferring context: 298 0.0s
=> CACHED [2/5] RUN pip install pika 0.0s
=> CACHED [3/5] WORKDIR /app 0.0s
=> CACHED [4/5] COPY chat.py . 0.0s
=> CACHED [5/5] RUN mkdir -p /app/data 0.0s
=> exporting to image 0.3s
=> => exporting layers 0.0s
=> => exporting manifest sha256:a1fe7e42d20a6275309c8895927a7063c0cbac3d83349b17a24410cffa9f4875 0.0s
=> => exporting config sha256:2c7e62ffe796b83822fed2588924353b5482aa7e1620221b4341ff4a00246216 0.0s
=> => exporting attestation manifest sha256:6c08c5086f1499e5c84a2e25b8723ecc2e7d0899e419f39ab2a092bcd27dcda 0.1s
=> => exporting manifest list sha256:d72941d7c801e79adc62356162f0f736dabeb8ce1da8fa2d7b19eeef49eca66b4 0.0s
=> => naming to docker.io/library/docker-advanced-lab-user-b:latest 0.0s
=> => unpacking to docker.io/library/docker-advanced-lab-user-b:latest 0.0s
=> resolving provenance for metadata file 0.0s
Image docker-advanced-lab-user-b Built
Container docker-advanced-lab-user-b-run-87ec7ea25eb3 Creating
Container docker-advanced-lab-user-b-run-87ec7ea25eb3 Created

--- No Previous Chat History ---

--- Chat Started (user-b connecting to rabbitmq) ---
Type message:
Received from user-a: hello
Type message: wazzzzzz
Sent by user-b: wazzzzzz
Type message:
Received from user-a: wazzzzzaaaa
Type message:
```

SS11:

```
[akshay@AKSHAY ~]~/mnt/c/Users/AKSHAY_AG/docker-advanced-lab
$ docker compose run --rm user-a
WARN[0000] /mnt/c/Users/AKSHAY_AG/docker-advanced-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid
potential confusion
WARN[0000] No services to build
[+] 1/1t 1/11
✔ Container docker-advanced-lab-rabbitmq-1 Running 0.0s
WARN[0000] No services to build
Container docker-advanced-lab-user-a-run-11a7c35f1167 Creating
Container docker-advanced-lab-user-a-run-11a7c35f1167 Created

--- Chat History ---
Sent by user-b: hello
Received from user-b: hello
Sent by user-a: wazzzzzaaaaa
Received from user-a: wazzzzzaaaaa
Sent by user-b: wazzzzzzzaaaaaaaaa
Received from user-b: wazzzzzzzaaaaaaaaa
--- End of History ---

--- Chat Started (user-a connecting to rabbitmq) ---
Type message:
Received from user-b: hello
Type message: hello friend
Sent by user-a: hello friend
Type message:
Received from user-b: wow
Type message: wow!!
Sent by user-a: wow!!
Type message: |
```

SS12:

```
(akshay@AKSHAY) - [ /mnt/c/Users/AKSHAY_AG/docker-advanced-lab ]
$ docker compose run --rm user-b
WARN[0000] /mnt/c/Users/AKSHAY_AG/docker-advanced-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid
potential confusion
WARN[0000] No services to build
[+] 1/1t 1/11
✓ Container docker-advanced-lab-rabbitmq-1 Running 0.0s
WARN[0000] No services to build
Container docker-advanced-lab-user-b-run-ee573d926b08 Creating
Container docker-advanced-lab-user-b-run-ee573d926b08 Created

--- Chat History ---
Sent by user-b: hello
Received from user-b: hello
Sent by user-a: wazzzzaaaaa
Received from user-a: wazzzzaaaaa
Sent by user-b: wazzzzzaaaaaaaaa
Received from user-b: wazzzzzaaaaaaaaa
--- End of History ---

--- Chat Started (user-b connecting to rabbitmq) ---
Type message: hello
Sent by user-b: hello
Type message:
Received from user-a: hello friend
Type message: wow
Sent by user-b: wow
Type message:
Received from user-a: wow!!
Type message: |
```